

Exam: AppStat 25/26

KU-ID: KVF851

Exam number: 6

Date(s): 15/01 to 16/01 2026

Note: In the following, all fits are done using iMinuit. A python script containing external functions is used and uploaded in appendix. Furthermore, for all parts where random number generation is used, the seed is fixed to 1 for reproducibility.

Problem I

1.1.1

Assuming the underlying PDFs for number of tails and hits are binomial, we can simulate 10000 trials for both. We obtain the following:

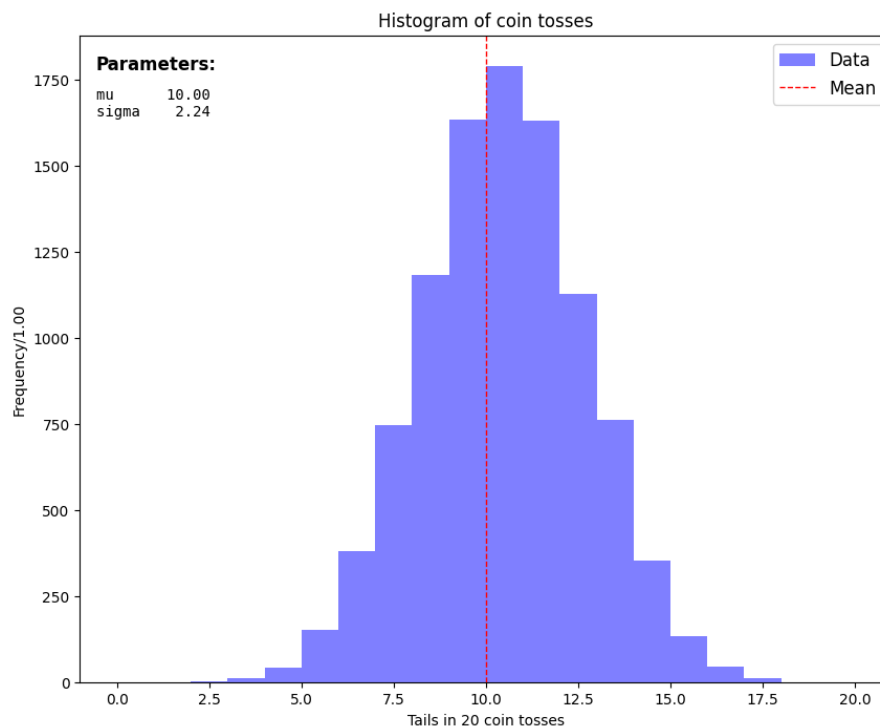


Figure 1: Distribution of amounts of tails.

Looking at both fig. 1 and fig. 2, it seems fair to approximate both distributions to be Gaussian.

Using the mean-value and standard deviations, we find a difference distribution that describes the difference in successes. For this distribution we get $\mu = 0$, and $\sigma = \sqrt{\sigma_c^2 + \sigma_g^2}$,

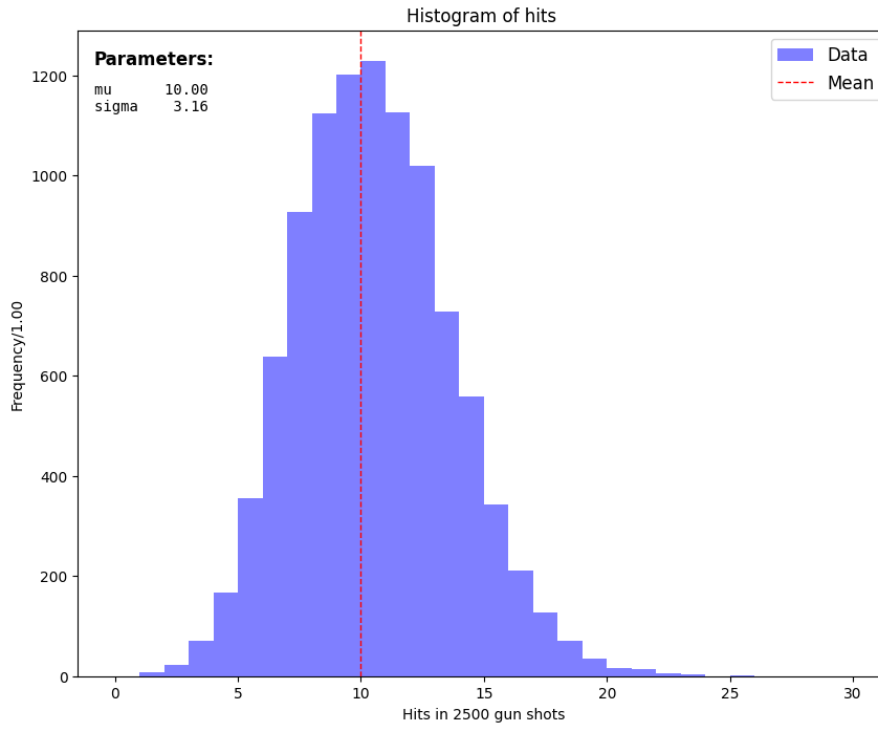


Figure 2: Distribution of amounts of hits.

with c, g denoting coin and gun distributions.

We thus find the area above 0 to be 0.5, so $P(N_1 > N_2) = P(N_2 > N_1) = 0.5$.

1.1.2

This amount can be found by simply simulating a larger sample, and finding the amount between 25 and 250.

Simulating 1 million trials of 2500 gun shots, we obtain $P(25 \leq N_2 \leq 250) = 0.000043 = 4.3 \cdot 10^{-5}$. For N_1 , we see that $N_1 \geq 25$ is not in the sample space, i.e. $P(N_1 \geq 25) = 0$.

1.1.3

Using `stats.binom.pmf`, we get that $P(N_1 = 10) = 0.176$ and $P(N_2 = 20) = 0.0018$. Since the two instances are independent of each other, we get:

$$P(N_1 = 10 \wedge N_2 = 20) = P(N_1 = 10)P(N_2 = 20) = 0.00032 \quad (1)$$

For the reverse case, we get $P(N_1 = 20) = 0.000001 = 1 \cdot 10^{-6}$ and $P(N_2 = 10) = 0.125$, thus:

$$P(N_1 = 20 \wedge N_2 = 10) = P(N_1 = 20)P(N_2 = 10) = 0.00000012 \quad (2)$$

Problem II

2.1

2.1.1

To find the value on z and it's uncertainty, we simply error propagate using sqrt on eq. (4.12). We build a function which incorporates ρ .

Using sympy, we obtain the following expressions:

$$z = \frac{\cos(x^2)}{\log(xy)} \quad (3)$$

with $\log$ being the natural logarithm, NOT $\log_{10}$. This yields the error on z to be:

$$\sigma_z = \sqrt{-\frac{2\rho\sigma_x\sigma_y \left(-\frac{2x \sin(x^2)}{\log(xy)} - \frac{\cos(x^2)}{x \log(xy)^2} \right) \cos(x^2)}{y \log(xy)^2} + \sigma_x^2 \left(-\frac{2x \sin(x^2)}{\log(xy)} - \frac{\cos(x^2)}{x \log(xy)^2} \right)^2 + \frac{\sigma_y^2 \cos^2(x^2)}{y^2 \log(xy)^4}} \quad (4)$$
$$(5)$$

Setting $\rho = 0$, and plugging in $x = 1.71, \sigma_x = 0.05, y = 10.1, \sigma_y = 0.3$, we get:

$$z_{\rho=0} = -0.343 \pm 0.010 \quad (6)$$

2.1.2

With a correlation of $\rho = 0.87$, using eq. (4), we get:

$$z_{\rho=0.87} = -0.343 \pm 0.007 \quad (7)$$

with a bit of a smaller error compared to eq. (6).

2.2

2.2.1

Using numpy's unique function, we find that $T \in \{0.2^\circ, 2.3^\circ, 3.9^\circ, 5.9^\circ, 7.8^\circ\}$. Sorting our densities w.r.t these temperatures, we find the mean, and its error, assuming an underlying gaussian PDF i.e. standard error on the mean, presented in table 1.

2.2.2

To argue for any outliers, we first plot the measurements in a scatter plot (fig. 3):

Temp.	Mean density in g/cm^3
0.2°	0.999855 ± 0.000003
2.3°	0.999952 ± 0.000003
3.9°	0.999972 ± 0.000003
5.9°	0.999951 ± 0.000004
7.8°	0.999865 ± 0.000004

Table 1: Mean densities and their errors for each unique temperature.

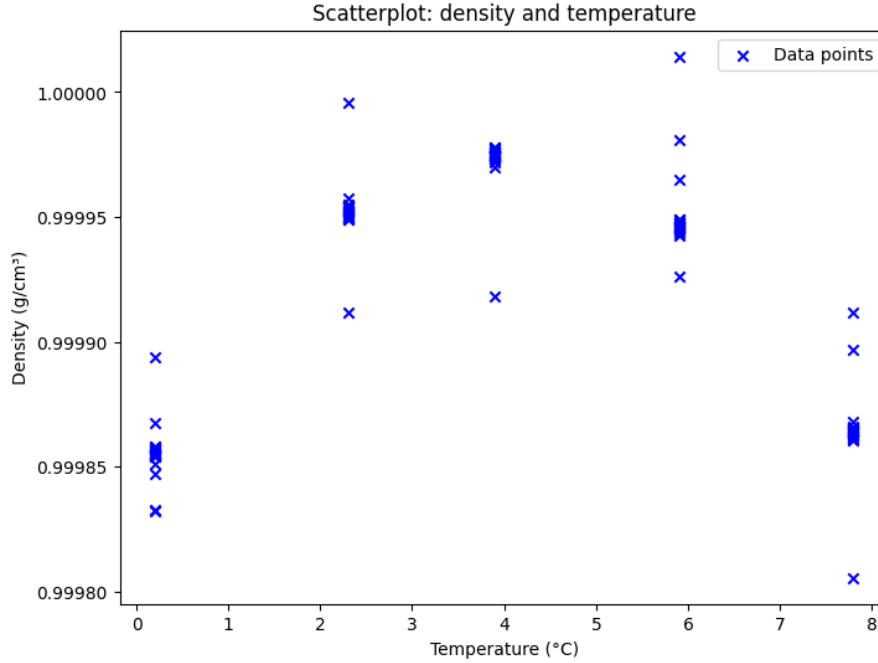


Figure 3: Scatter plot of density measurements at different temperatures.

Looking at fig. 3, it is immediately obvious that for each temperature, there are outliers that must be unlikely. Assuming the measurements are distributed w.r.t. Gauss, with a significant level of about 1%, all measurements more than 3σ must be excluded. Let us remove all measurements 3σ away from the mean for each unique temperature, and recalculate.

After removing the outliers, the new means and their errors are calculated in the same way as described above. The new means and errors, and the removed data points, can be seen in table 2.

Temp.	Cleaned mean density in g/cm^3	Removed data
0.2°	0.9998532 ± 0.0000019	{0.9998939}
2.3°	0.999950 ± 0.000002	{0.999996}
3.9°	0.9999749 ± 0.0000005	{0.9999182}
5.9°	0.999948 ± 0.000002	{1.0000143}
7.8°	0.999868 ± 0.000003	{0.9998054}

Table 2: Mean densities and their errors after cleaning the data, with the removed data-points at a 3σ -level.

On fig. 4, we see a scatter plot showcasing the accepted and rejected data points.

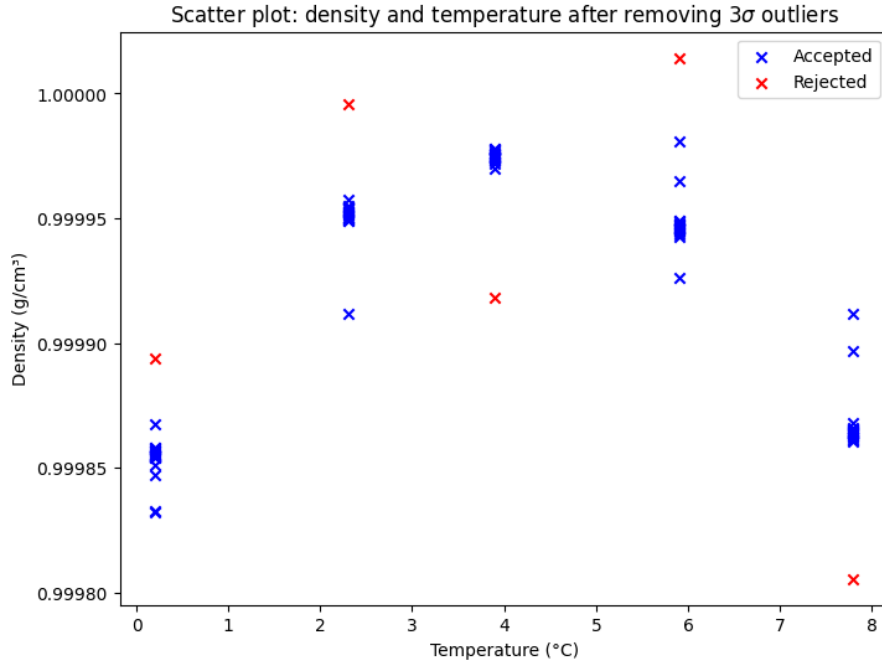


Figure 4: Scatterplot of density measurements at unique temperatures with outliers marked as red x's.

2.2.3

The temperature with the highest density is found to be 3.9° , with a density of:

$$\rho_{3.9^\circ} = (0.9999749 \pm 0.0000005) g/cm^3 \quad (8)$$

To see how confident we are in the fact that this really is the highest density, let us calculate how many σ_{comb} away the second highest density is. We see that:

$$\sigma_{comb} = \sqrt{\sigma_{3.9^\circ}^2 + \sigma_{2.3^\circ}^2} \quad (9)$$

is the combined error added in quadrature. We find that the second highest density is:

$$\rho_{2.3^\circ} = (0.999950 \pm 0.000002) g/cm^3 \quad (10)$$

at 2.3° .

Comparing the two we find $\rho_{3.9^\circ}$ is $11.6\sigma_{comb}$ away from $\rho_{2.3^\circ}$. We can thus be very(!) confident that this is the temperature with the highest density i 3.9° .

2.2.4

Looking at fig. 4, we assume the density evolves as a 2nd degree polynomial w.r.t. temperature.

Doing a χ^2 -fit with:

$$f(x) = ax^2 + bx + c \quad (11)$$

we obtain the following fit (fig. 5):

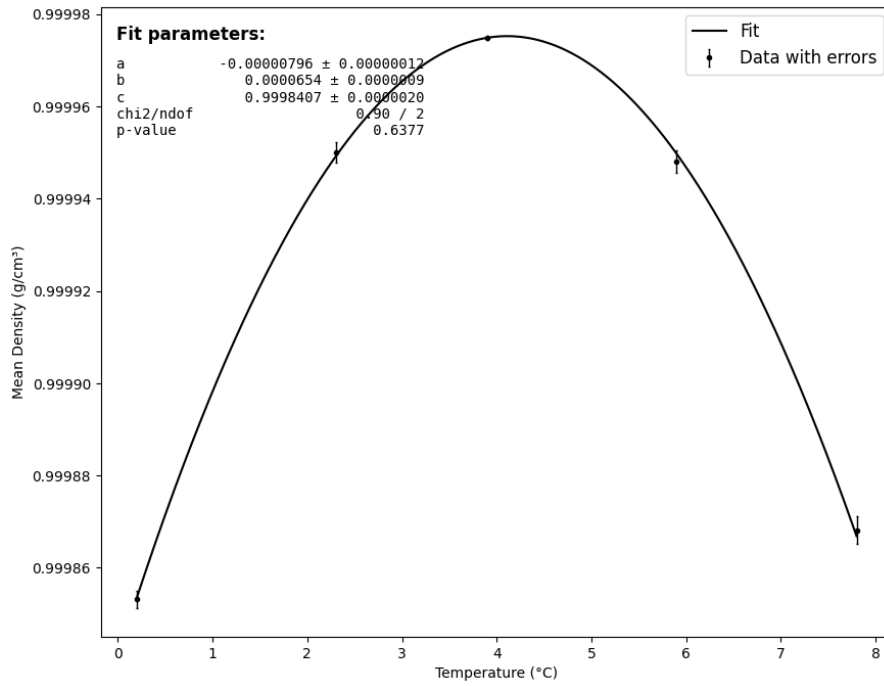


Figure 5: Polynomial fit to density vs. temperature data, with χ^2 , Ndof and p-value.

The fit parameters are summarized in table 3.

Parameters	Fit value
a	$-0.00000796 \pm 0.00000012$
b	0.0000654 ± 0.0000009
c	0.9998407 ± 0.00000020
p(chi2 = 0.9, ndof = 2)	0.6377

Table 3: Summarizing fit parameters for fig. 5.

From the fit it is evident that a p-value of 0.6377 is quite good! The χ^2 -value is a bit low, about 1/2 of ndof. This must partly be due to the low statistics in this problem, and the fact that the fitting function has a relatively large amount of fit parameters w.r.t. the amount of data points.

We can find the temperature where the density is highest by utilizing the fact that the function is a second degree polynomial, i.e. the x-value for which the function is largest in y (or lowest) is $T_{max} = -b/2a$.

Using error propagation, we get:

$$\sigma_{T_{max}} = \sqrt{\left(\frac{-1}{2a}\sigma_b\right)^2 + \left(\frac{b}{2a^2}\sigma_a\right)^2} \quad (12)$$

This gives us a temperature where the density is highest to be:

$$T_{max} = (4.11 \pm 0.09)^\circ \quad (13)$$

This is about $2.47\sigma_{T_{max}}$ away from 3.9° . So the found maximum temperatures still agree within a significance level of 1%, but not 5%.

2.3

2.3.1

Using eq. (4.12) again with $\rho = 0, y = 0$, we get the following expression for the error on γ :

$$\sigma_\gamma = \sqrt{\frac{\beta^2 \sigma_\beta^2}{(1 - \beta^2)^3}} \quad (14)$$

Inserting in eq. (14), we get:

$$\gamma = 1.155 \pm 0.015 \quad (15)$$

2.3.2

Now if $\beta = 0.95 \pm 0.02$, using eq. (14), we would get:

$$\gamma = 3.2 \pm 0.6 \quad (16)$$

To test if the uncertainty is symmetric, we draw 1 million samples from a Gaussian distribution with mean $\beta = 0.95$ and standard deviation $\sigma = 0.02$, i.e. we assume the error on β to be Gaussian. We then calculate γ -values based on the samples and plot, see fig. 6.

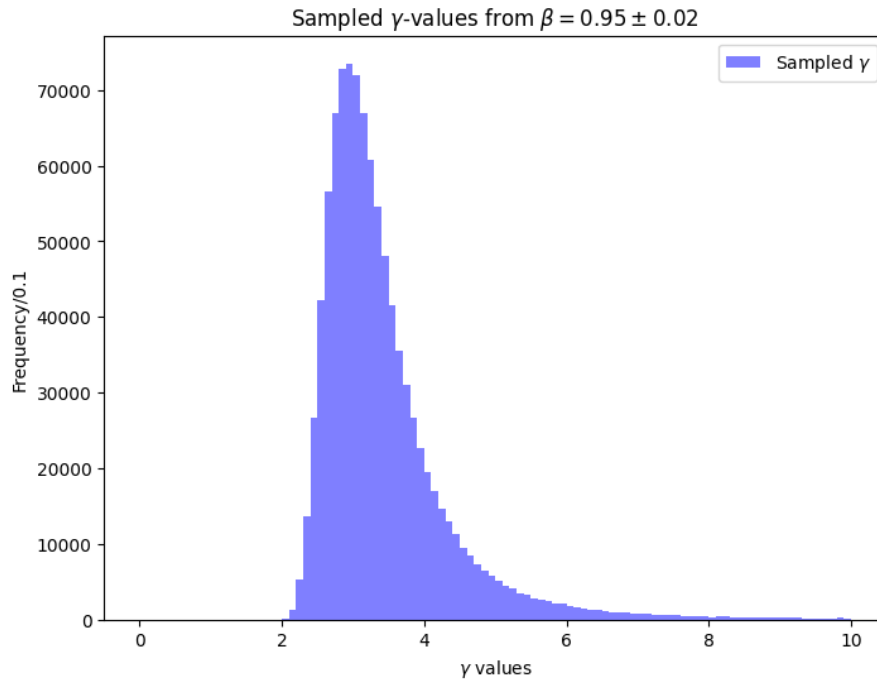


Figure 6: Distribution of γ -values from $\beta = (0.95 \pm 0.02)$

Looking at fig. 6, we see that the distribution is highly asymmetric, and thus will the error also be asymmetric.

Problem III

3.1

3.1.1

We extract values from $x = \sum_j^N \varepsilon_j / j$ by simply taking the sum over j , and incorporating the variability of ε_j by using numpy's choice function, with the sample space $\{-1, 1\}$ and equal probability.

Extracting 10000 values for $N = 25$, we get the following distribution (fig. 7):

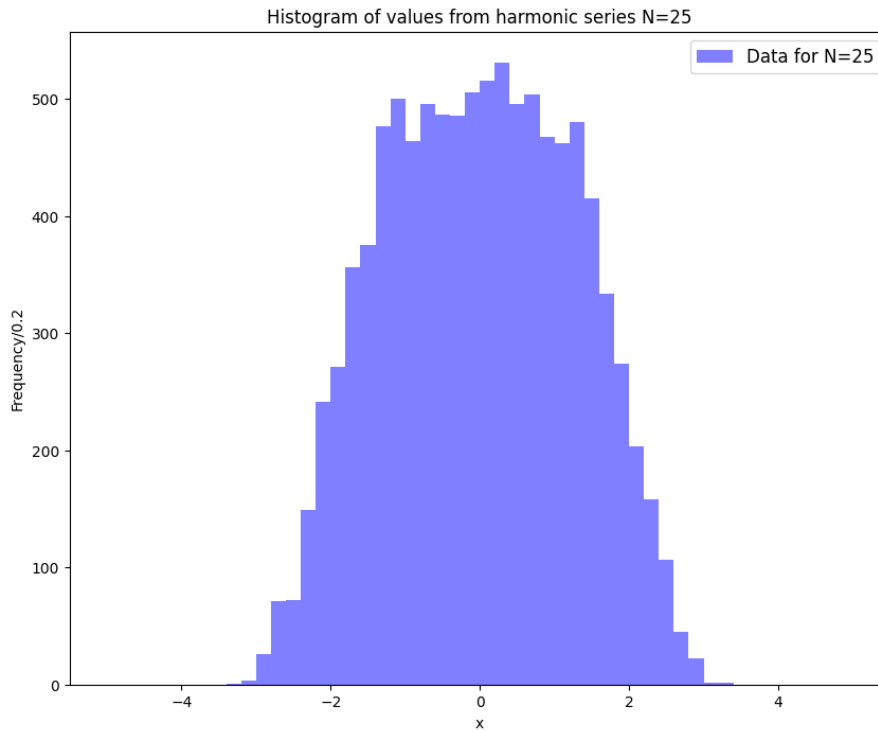


Figure 7: Distribution of x -values from the harmonic series for $N=25$.

As can be seen, the figure looks pretty symmetric.

3.1.2

Let us test if the distribution is symmetric around 0, by extracting the positive and negative values. The amount of positive and negative values should be equal. To test this we do a frequency test, i.e. we determine Pearson's χ^2 , choose a null-hypothesis that states that the distributions are the same and choose a significant level of 5%.

The results are presented on fig. 8.

On fig. 8 it is evident that the frequency test results in a p-value of 0.704, and we thus accept our null-hypothesis, i.e. the distribution is symmetric around $x=0$.

We may also take the absolute value of the negative values, and compare these with the positive values in a KS-test. The results are seen on fig. 9.

Looking at fig. 9, we see a p-value of $p = 0.279$. This is above the chosen significant level, so we accept our null hypothesis, i.e. the distribution of the harmonic series for $N=25$ is symmetric around 0.

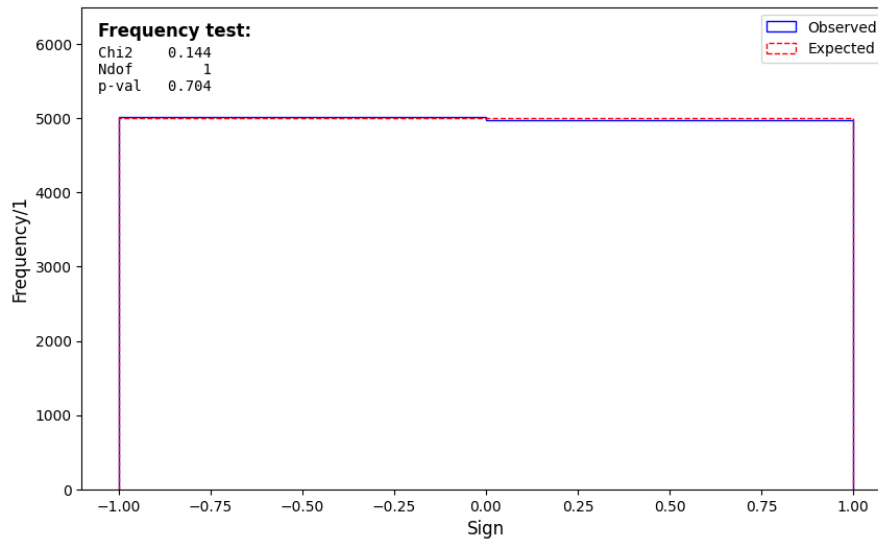


Figure 8: Frequency test of the sign of values extracted from the harmonic series to test if the distribution is symmetric around $x=0$.

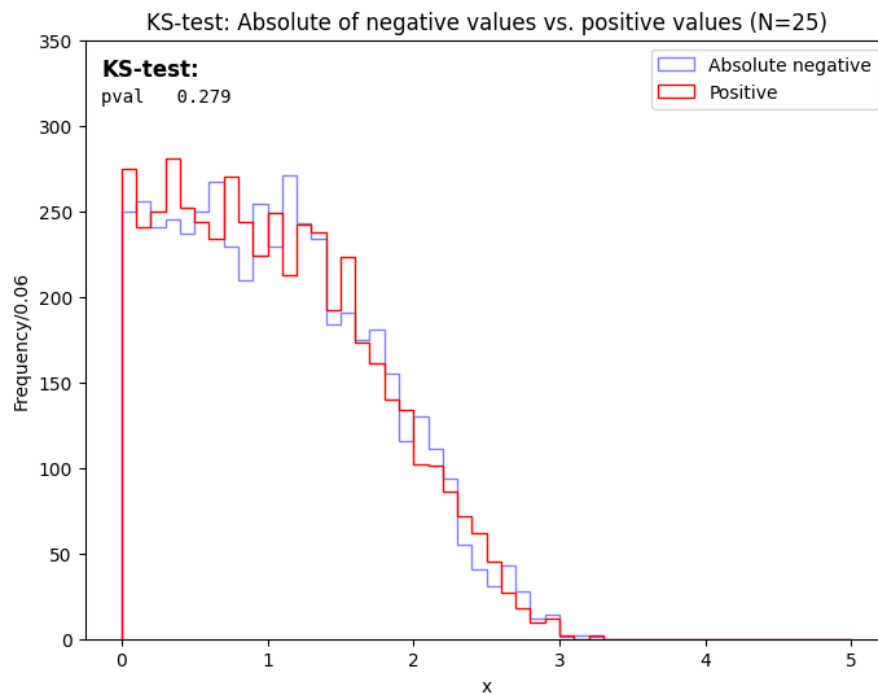


Figure 9: KS-test: Absolute of negative values vs. positive values from the harmonic distribution for $N=25$.

3.2.3

Let simulate the x -values for $N = 250$ and plot the distribution (fig. 10):

To test if they really are similar, we do a Kolmogorov-Smirnoff (KS) test. With a null hypothesis that they are the same, we choose a significant level of 5%.

The results are plotted in fig. 11.

We see that the p -value of $p = 0.523$ exceeds our significant level. We thus accept our null-hypothesis, and conclude that they are consistent with each other.

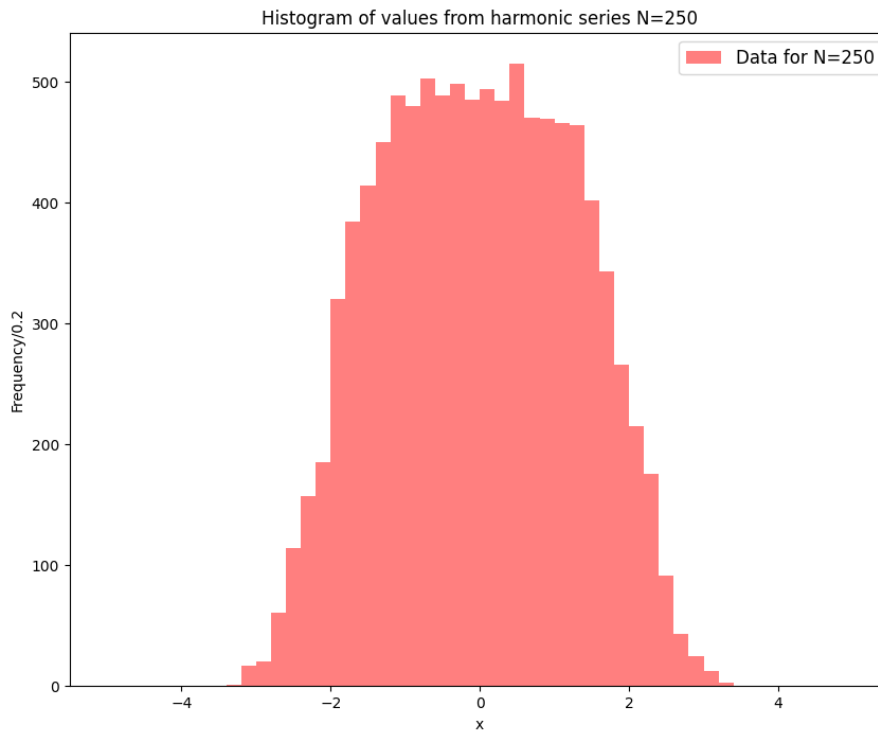


Figure 10: Distribution of x -values from the harmonic series for $N=250$.

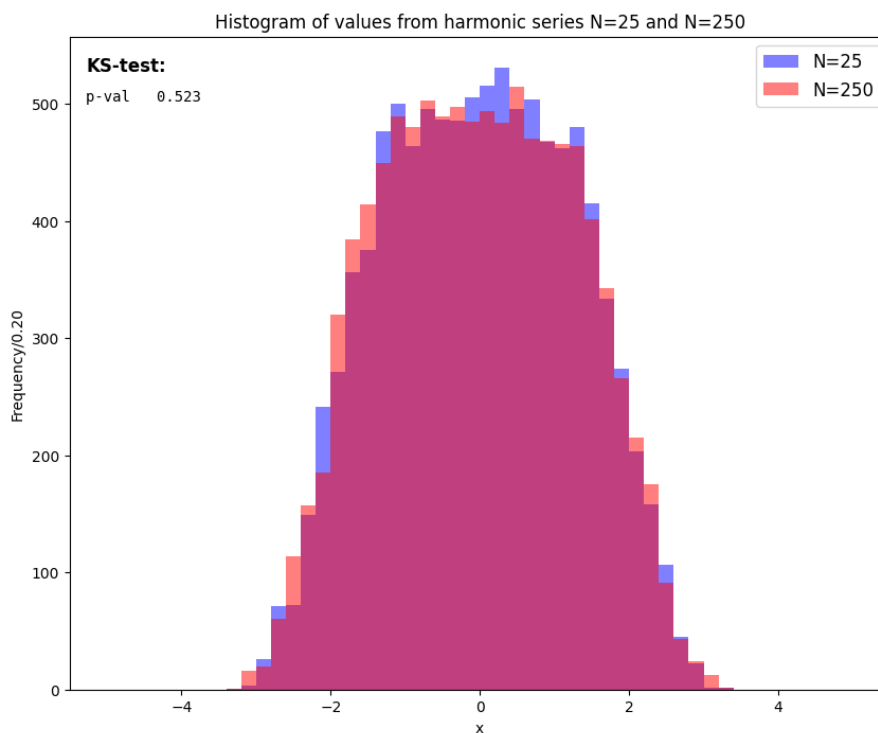


Figure 11: KS-test for distributions of x for $N = 25$ and $N = 250$. Poisson errors are omitted for visualization.

3.2.4

Finding the maximum PDF value for fig. 10, by dividing the maximal frequency by bin width and data-points (i.e. normalizing), and doing the same for the Poisson error, we get a maximum PDF value from the distribution to be:

$$x_{PDF_{max}} = 0.26 \pm 0.1 \quad (17)$$

Choosing a null hypothesis that they are the same, with a significant level of 5%, and doing a two-tailed z-test against $1/4$, we get at z-value of $z = 0.66$ with $p = 0.5086$. We must then accept our null-hypothesis and claim that the maximum PDF value is consistent with $1/4$.

Problem IV

4.1

4.1.1

To start of, let us plot the H_0 distribution:

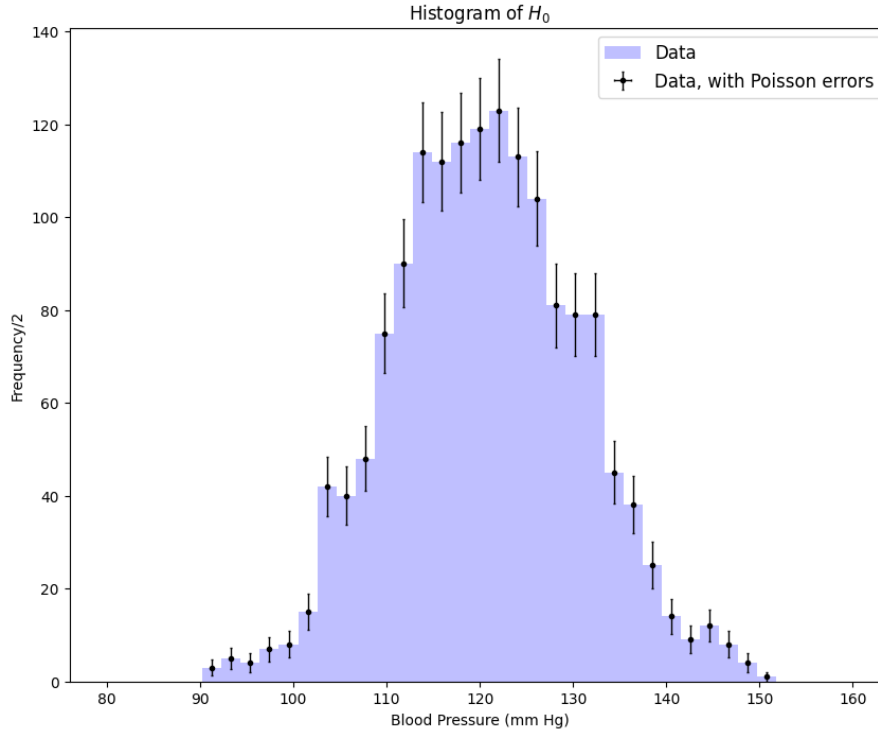


Figure 12: Histogram of blood pressure for healthy patients.

We are in the case with no given errors on our measurements, and want to estimate a 95% confidence interval for $\bar{s}$. Looking at fig. 12, the distribution is approximately Gaussian. But, we have no prior knowledge of the resolution, so we may find the 95% confidence interval on $\bar{s}$ using Students t. But since it converges to the normal distribution for large n , we will also estimate it using the standard 95% interval, i.e. $\pm 1.96\sigma$.

The mean is determined to be $\bar{s} = (120.7 \pm 0.3)$ mmHg, with the error from the standard error on the mean. The 95% confidence interval using Students t is found to be $[120.2, 121.2]$ mmHg, and using the Gaussian approximation we get the same, i.e. $[120.2, 121.2]$ mmHg.

So the 95% confidence interval is $[120.2, 121.2]$ mmHg.

4.1.2

Our approach is the following. The separation metric $d = |SBP - \bar{s}|$ calculates the distance between each SBP and the estimated mean SBP for healthy(!) patients. We must then suspect that $d_{healthy} < d_{non-healthy}$, since non-healthy patients have assumed different blood pressures than healthy. This means as we increase d , we should classify more patients as sick.

Plotting d -scores for healthy and non-healthy patients we see:

Looking at fig. 13, we see the above mentioned trend is followed.

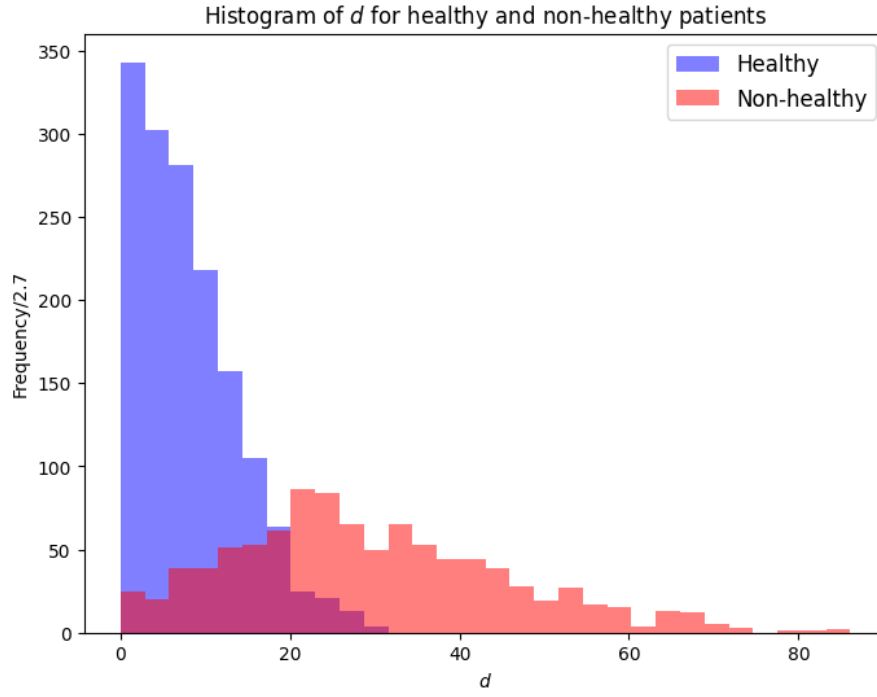


Figure 13: Distribution of test statistic d for healthy and non-healthy patients. Poisson errors omitted for visualization.

We determine the true positive rate (TPR) and false positive rate (FPR) using:

$$TPR = \frac{TP}{TP + FN} \quad (18)$$

$$FPR = \frac{FP}{FP + TN} \quad (19)$$

with TP being true positive, FN false negative, FP false positive and TN true negative.

We sort our d -scores in descending order, i.e. first entry is highest, and for each sorted d -value we calculate TP, FN, FP and TN, and then use eq. (18).

Doing this, we get the following ROC curve for d (fig. 14):

Looking at fig. 14, we see that d is a pretty good separation metric, since the ROC curve bunches up to the left top corner, i.e. we get a good TPR.

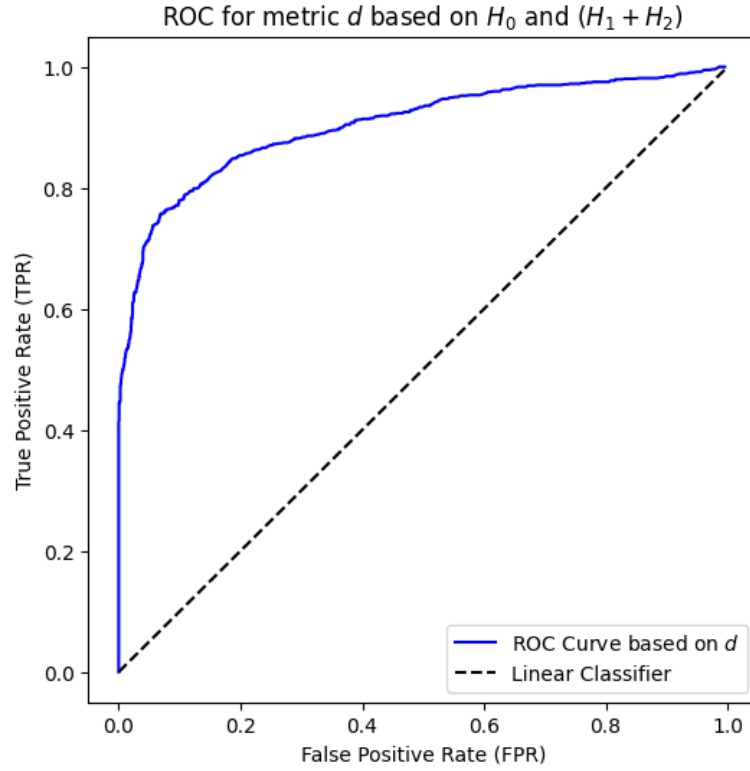


Figure 14: ROC curve based on separation metric d for healthy and non-healthy patients.

4.1.3

We'll use Bayes theorem, so:

$$P(H_2|SPB = 98) = \frac{P(SPB = 98|H_2)P(H_2)}{P(SPB = 98)} \quad (20)$$

Assuming all priors are equal, we know $P(H_0) = P(H_1) = P(H_2) = 1/3$. Assuming the underlying blood pressure distribution is Gaussian for H_2 , we find $P(SPB = 98|H_2)$ using the mean blood pressure and its standard deviation, and then simply using stats' norm.pdf function. We obtain $P(SPB = 98|H_2) = 0.061$. We find $P(SPB = 98)$ by using:

$$P(SPB = 98) = P(SPB = 98|H_0)P(H_0) + P(SPB = 98|H_1)P(H_1) + P(SPB = 98|H_2)P(H_2) \quad (21)$$

The approach for each likelihood is the same as for $P(SPB = 98|H_2)$, and knowing all the priors, eq. (21) is done easily.

We get $P(SPB = 98|H_0) = 0.003$ and $P(SPB = 98|H_1) = 0.00049$.

Putting this together, we get:

$$P(H_2|SPB = 98) = 0.945 \quad (22)$$

Looking at the exercise it may look like only priors for H_0 and H_2 are assumed equal.

Assuming $P(H_0) = P(H_2) = 0.5$ and forgetting about $P(H_1)$, we get:

$$P(H_2|SPB = 98) = \frac{P(SPB = 98|H_2)}{P(SPB = 98|H_0) + P(SPB = 98|H_2)} \quad (23)$$

since $P(H_1) = 0$ and the priors drop out. Using eq. (23), we get:

$$P(H_2|SPB = 98) = 0.9516 \quad (24)$$

i.e. not a big change!

4.1.4

Using the data, we get find the priors by simply dividing the amount of data points for each prior with the total data set. We get $P(H_0) = 0.6137$, $P(H_1) = 0.3151$, $P(H_2) = 0.0713$. Inserting in eq. (20), we get:

$$P(H_2|SPB = 98) = 0.679 \quad (25)$$

which is remarkably smaller than the above.

4.1.5

Plotting both the healthy, non-healthy and all patients in a histogram, we get the following (fig. 15):

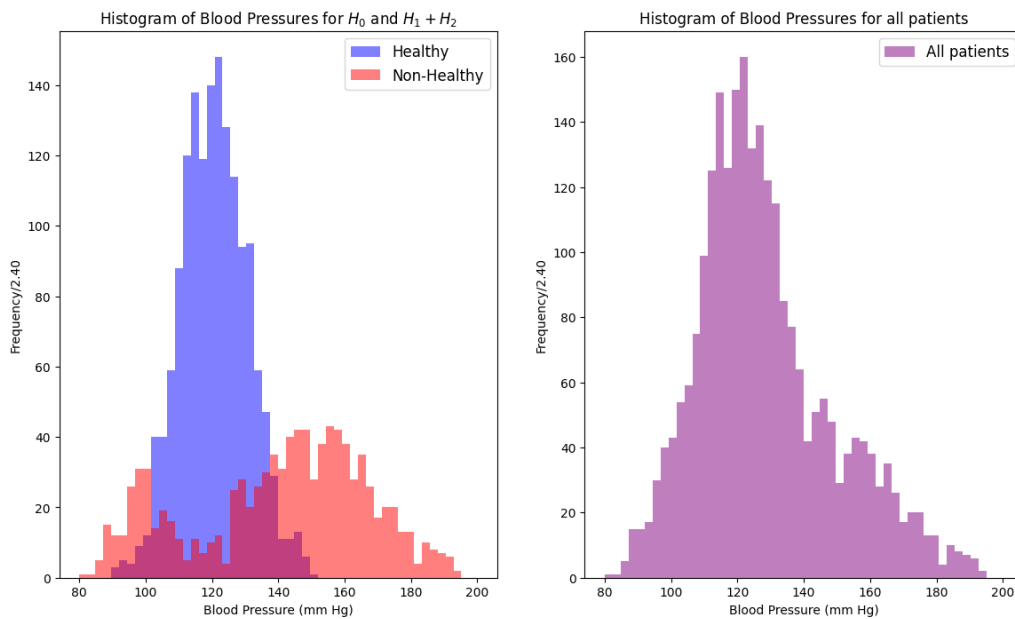


Figure 15: Histogram showcasing distribution of healthy and non-healthy patients (left) and all patients (right).

Looking at the left at fig. 15), it is evident that we can fit 3 gauss for each group. But looking at the right on the same figure, the entire distribution may be approximated by a double gauss fit. So let us try that.

Doing a χ^2 -fit, we get the following (fig. 16):

Looking at fig. 16, we see that the double Gaussian fit fits well. We get a $\chi^2/Ndof = 55.14/43$ and a p-value of $p = 0.1015$, above any reasonable significant level. I unfortunately don't have time to fit more PDFs.

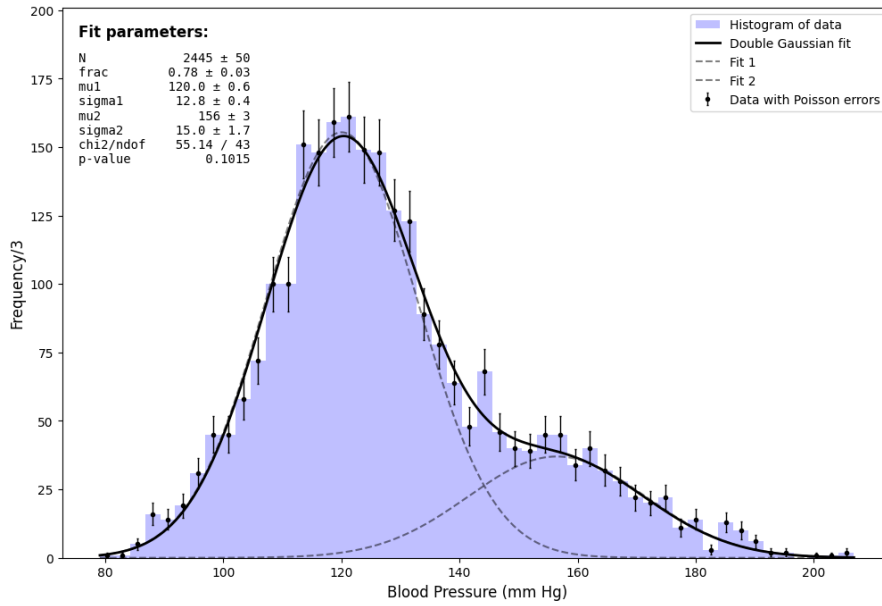


Figure 16: Double gauss χ^2 -fit on distribution of blood pressure for all patients.

Problem V

5.1

5.1.1

To fit according to the function $f_0(t) = vt$, where we've only been given errors on t , we'll need to first error propagate.

Using eq. (4.12) (assuming $\rho = 0$ and assuming v is constant), we get:

$$\sigma_{f_0} = v\sigma_t \quad (26)$$

i.e. we need to know v to get errors on $f_0(t)$. So first, we'll forget about σ_t , add dummy errors on d , do a fit and using the fit parameters, do a new fit with the propagated errors.

Setting an initial error of $\sigma_d = 50$ and doing an initial fit we get the fit shown in fig. 17.

Which is an OK fit looking at the p-value of $p = 0.9277$ and a $\chi^2/\text{ndof} = 11.65/20$. It is obvious that the errors are overestimated.

The fit yielded $v = (6.80 \pm 0.08) \text{ m/s}$.

The refit result can seen on fig. 18.

Looking at fig. 18, it is evident that the model fails. The p-value is very small, $p < 10^{-10}$ and $\chi^2/\text{ndof} = 365.71/20$ is very large.

The linear fit model fails.

5.1.2

Looking at fig. 18, the data might be modelled using a standard third degree polynomial function without a constat, since the persons best times might be slower the longer they go. The fit is on fig. 19.

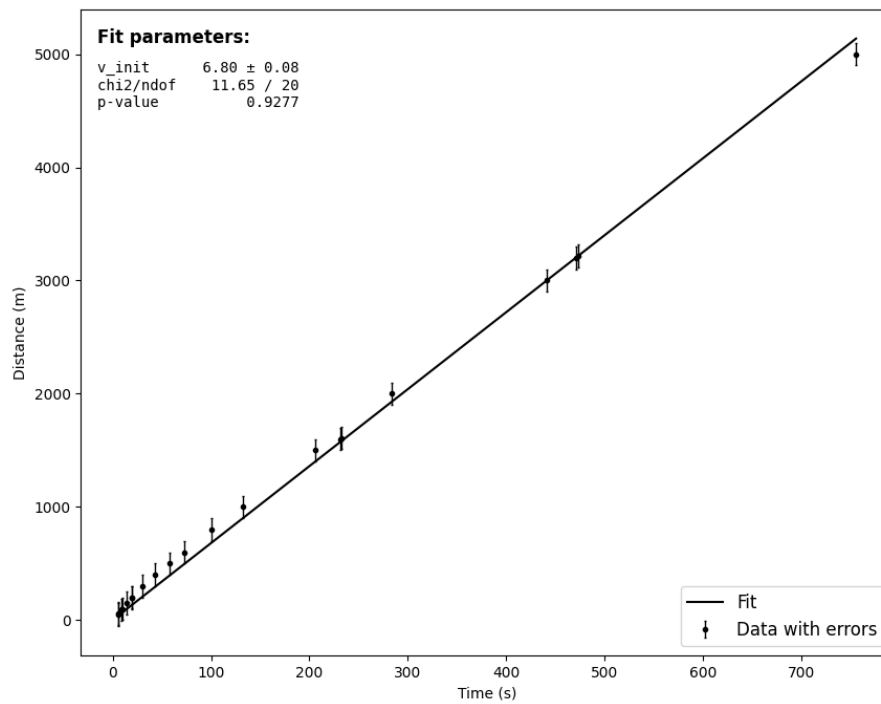


Figure 17: d vs t and initial fit with dummy errors.

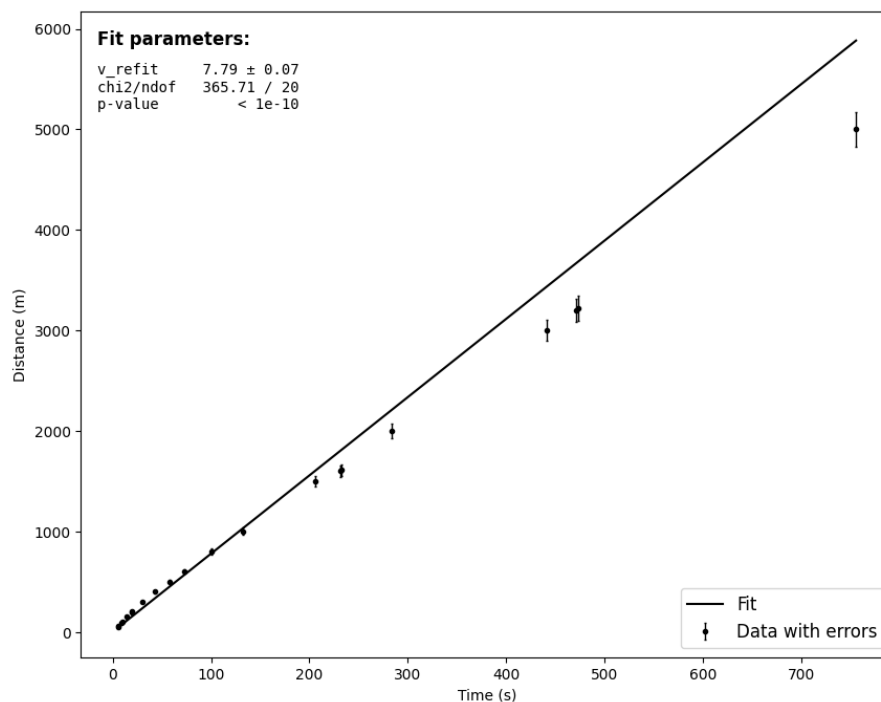


Figure 18: Refit of d vs t to f_0 .

Immediately we see a pretty poor p-value, and a χ^2 about 3x ndof. I tried to do several fits, but I've run out of time... It must be stated that we got errors on y using error propagation in the same way we found errors on y for the linear fit, i.e. doing an initial fit, and then refitting using found fit parameters and their errors. This is WRONG! But no time to fix...

If the running times follow the third degree polynomial model, the most likely record to be broken is the 5000m run. This was concluded by looking at the residuals, and determining that this was the data point with the largest negative residual w.r.t. the model. The residual

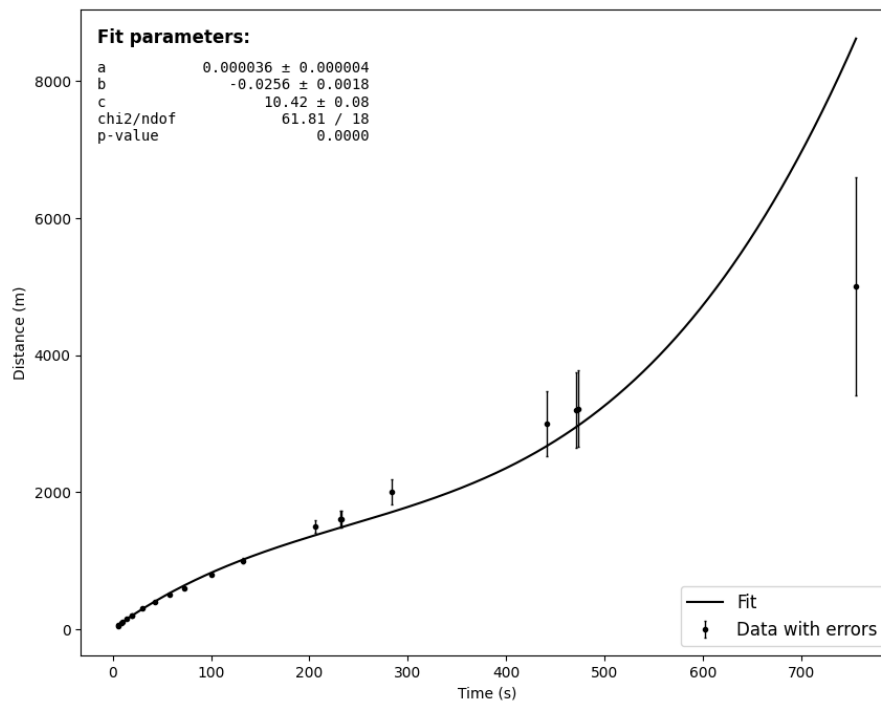


Figure 19: d vs t for a third degree polynomial fit with no constant.

plot can be seen on fig. 20.

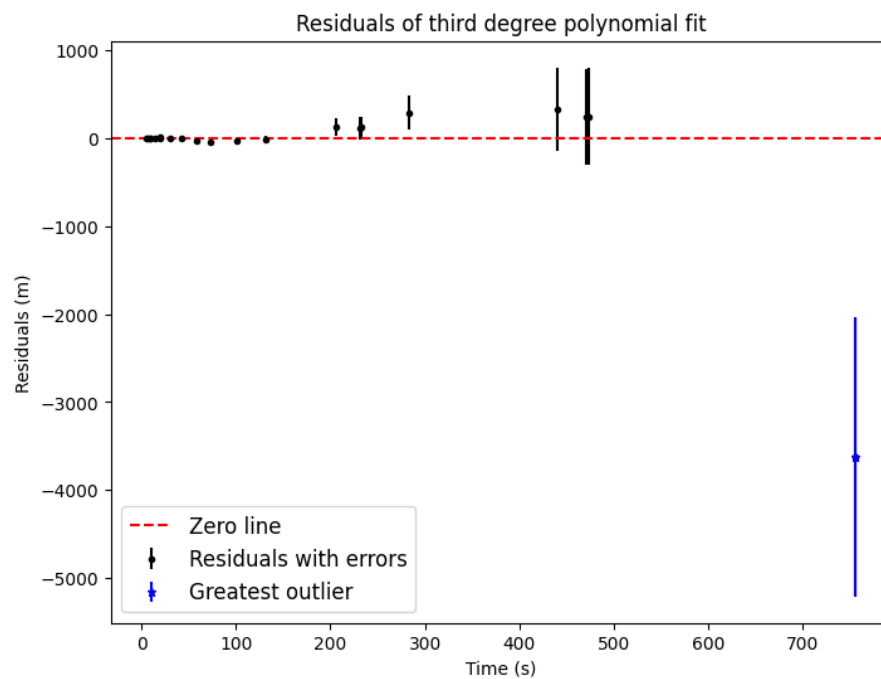


Figure 20: Residual plot using the same model and fit parameters as in fig. 19.

A large negative residual indicates that the persons record at the distance is significantly lower than what would be expected.

5.2

5.2.1

Let us start by plotting the frequency by itself:

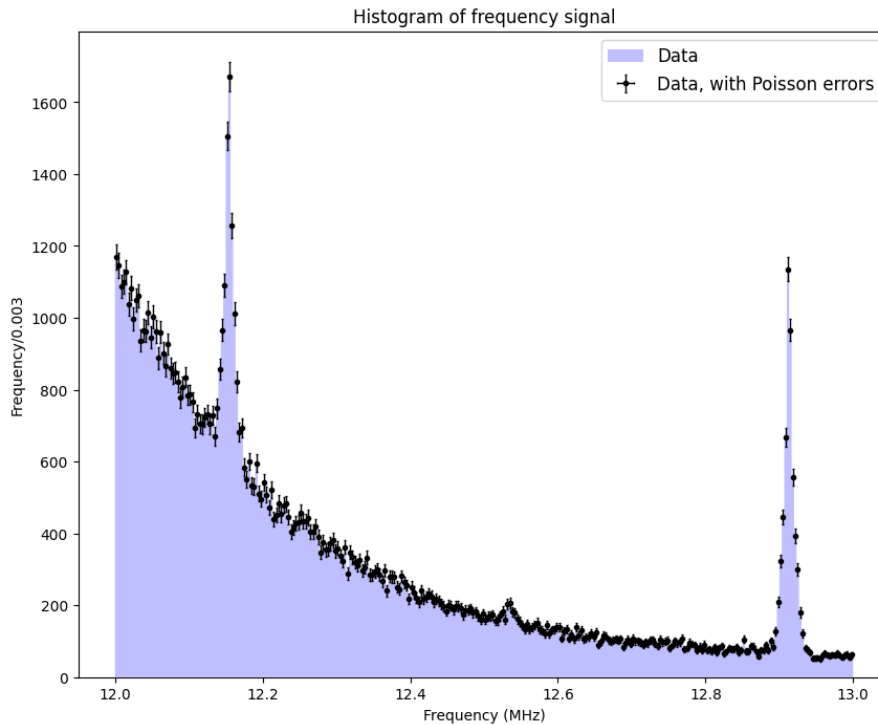


Figure 21: Histogram of frequency data

Looking at fig. 21 we see two clear peaks at frequencies around 12.15 and 12.95. The background looks overall to be a decaying exponential.

From doing an enormous amount of fitting to these peaks, we've learned that the model is very(!) sensitive to start guesses. So the approach is the following.

We fit the peaks twice using two different models, but with the same exponential background. The first model uses a standard Gaussian to fit the peak. The second model uses a Cauchy distribution (see p. 46 Barlow) to fit the peak, since the peak is very sharp.

First we fit the background using:

$$f(x)_{bkg} = \frac{1}{\tau} \exp\left(-\frac{(x - x_0)}{\tau}\right) \quad (27)$$

with τ characterizing the decay, and x_0 being an off-set, i.e. the decay-point.

Removing both peaks and fitting

Let us remove the peaks and fit the background with eq. (27), we get the following fit:

Looking at fig. 22, it is evident that the fit is not amazing; the χ^2 -value is enormous and the p-value is very small, probably due to the fact that the model doesn't actually describe the data, the missing points and possibly the small peak. But! It does give us an idea of the parameters of eq. (27). We get:

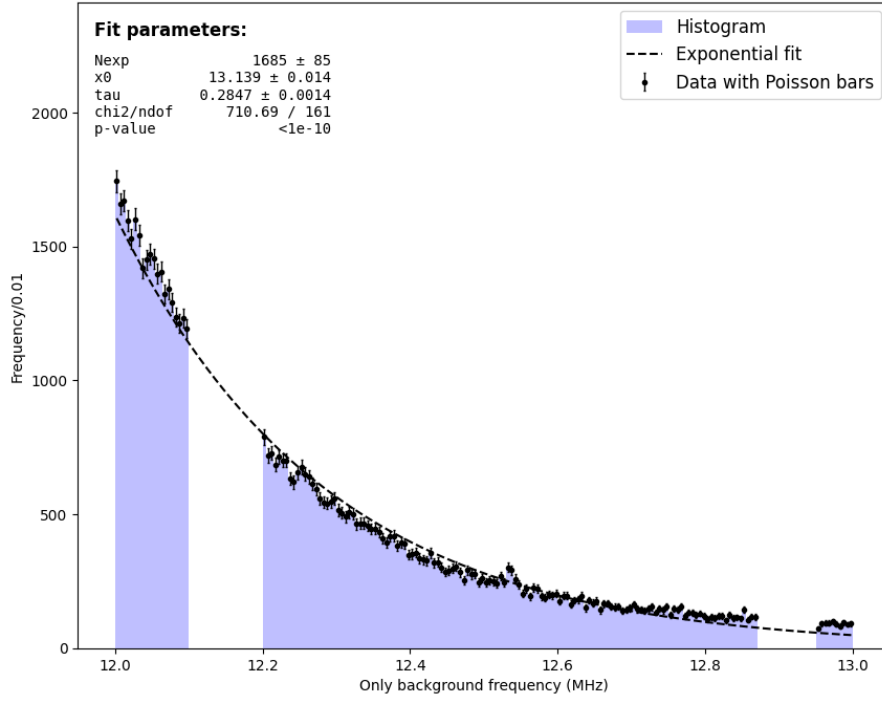


Figure 22: Exponential fit on background frequencies.

$$x_0 = 13.139 \pm 0.14 \quad (28)$$

$$\tau = 0.2847 \pm 0.0014 \quad (29)$$

Using eq. (28) as start guesses, we get for fitting peak 1 with a Gaussian and an exponential background (fig. 23):

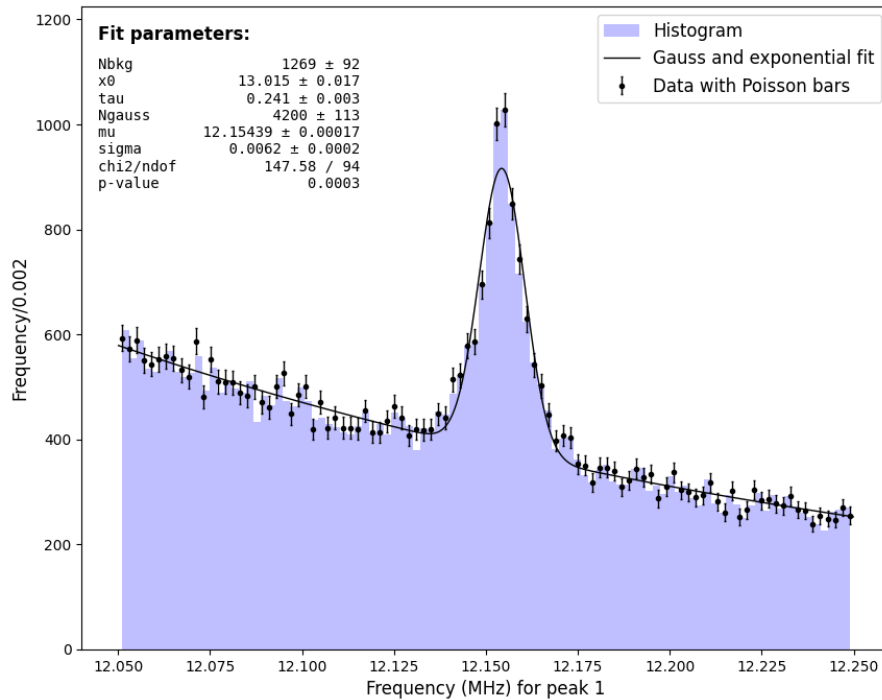


Figure 23: Fit of peak 1 using Gaussian model and exponential background.

Looking at fig. 23, we see an alright fit. The model doesn't really capture the sharpness of the peak. We do get a very small p-value of $p = 0.0003$ and a χ^2 which is about $3/2 \times \text{Ndof}$.

Using a Cauchy model to capture the peak instead, we get the following:

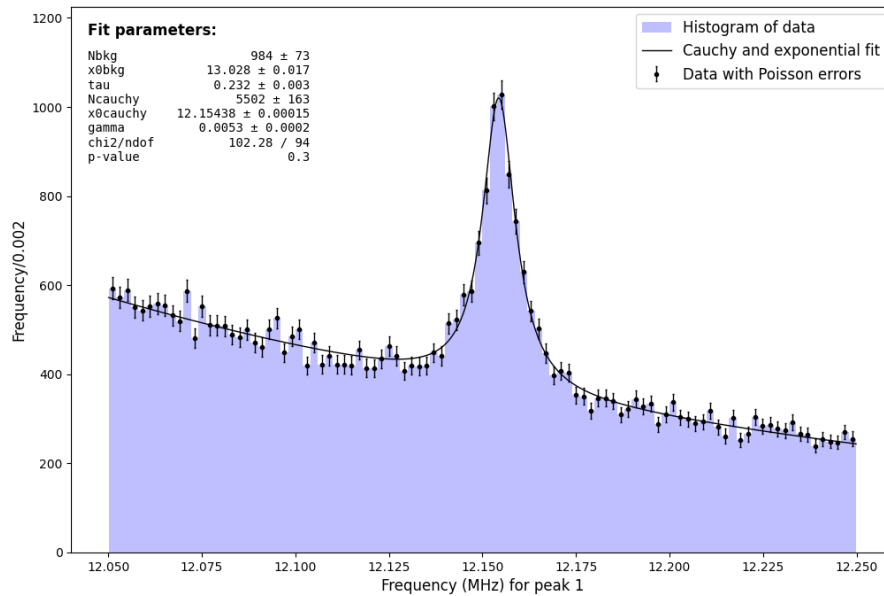


Figure 24: Fit of peak 1 using Cauchy model and exponential background.

Looking at fig. 24, we see a very(!) nice p-value of $p = 0.3$ and a $\chi^2/\text{ndof} = 102.28/94$, indeed a very good fit. It really captures the sharpness of the peak.

Now, let us focus on peak 2. Doing first a Gaussian fit with exponential background, we get:

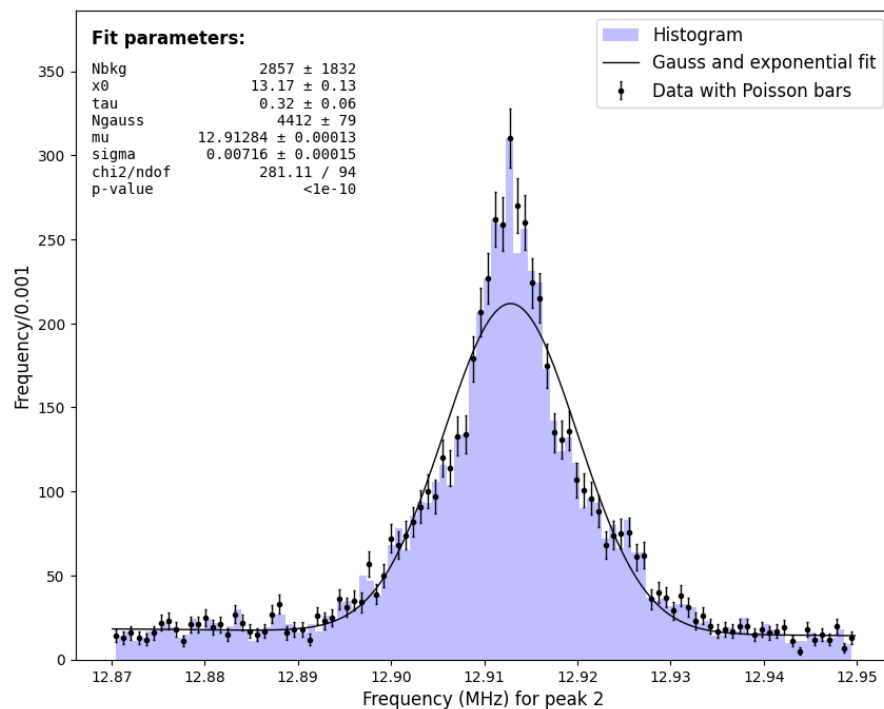


Figure 25: Fit of peak 2 using Gaussian model and exponential background.

Looking at fig. 25, we see a pretty poor fit. The χ^2 -value is very large and the p-value is extremely small. This is not a good fit model.

Using the Cauchy model, we get:

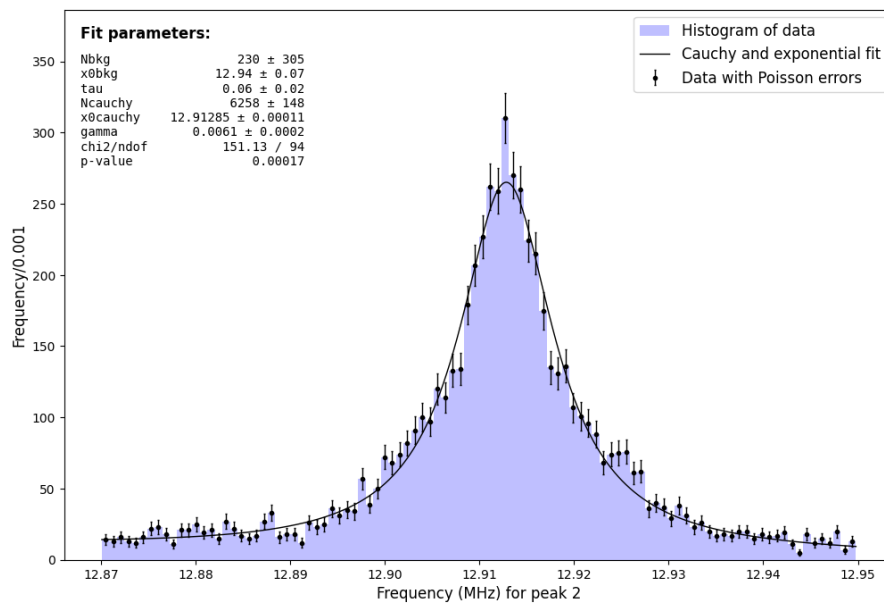


Figure 26: Caption

Looking at fig. 26, we see a better χ^2 and p-value compared to fig. 25, but the values are still off quite a bit.

This was unfortunately the best fits i could get. I also tried a double Gaussian fit with exponential background on both peaks, but they yielded slightly worse χ^2 and p-values.

For peak 1, the fit is:

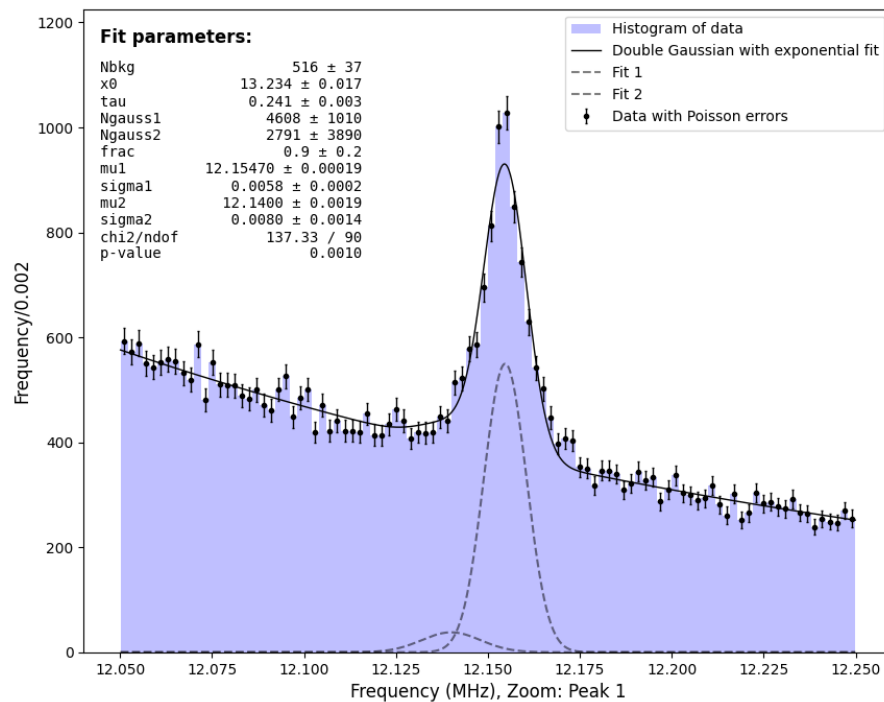


Figure 27: Double Gaussian fit with exponential background for peak 1.

Comparing to fig. 24, we see a much worse p-value and higher χ^2 . The fit on fig. 24 remains our best fit.

For peak 2, we get:

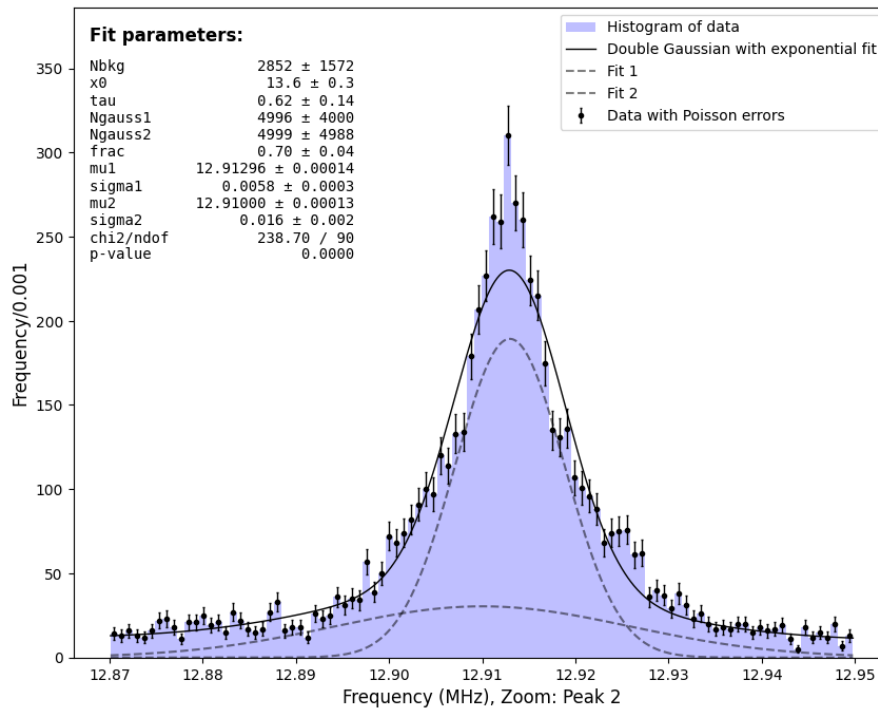


Figure 28: 7
Double Gaussian fit with exponential background for peak 2.

Upon comparing fig. 26, we again see a much smaller p-value and χ^2 . The fit on fig. 26 remains the best fit.

5.2.2

Above we have seen that single Gaussian fits are NOT consistent with the peaks, since their χ^2 values are a lot bigger than Ndof, and their p-values are smaller than any reasonable significant level.

To compare if the two peaks are consistent with each other, we do a two sample z-test for the Gaussian fits from fig. 23 and fig. 25, where we compare the fitted standard deviations.

Upon comparing σ_{p1}, σ_{p2} , we get at z-value of $z = -0.18$ with a p-value of $p = 0.43$. We can thus conclude, that the two Gaussian fits are consistent with each other with a significant level of 5%.

5.2.3

This part will be solved in two ways. The first approach is to sort the D and E data by removing all data points with a z-score of more than 2. The second approach is to use Fishers LDA. Using our eyes, we first categorize the given frequency into two boxes, signal and background, giving us two species. Using D and E as our two separation variables, we calculate the fisher weights and their fisher scores. We then do a cut, such that we only keep the best guesses for signal, i.e. we choose the top 90%.

z-score approach

For the first approach, removing all points where the respective z-score for both D and E are 2 or larger, we remove 6202 data points.

Plotting the original against the z-score cleaned data, we get:

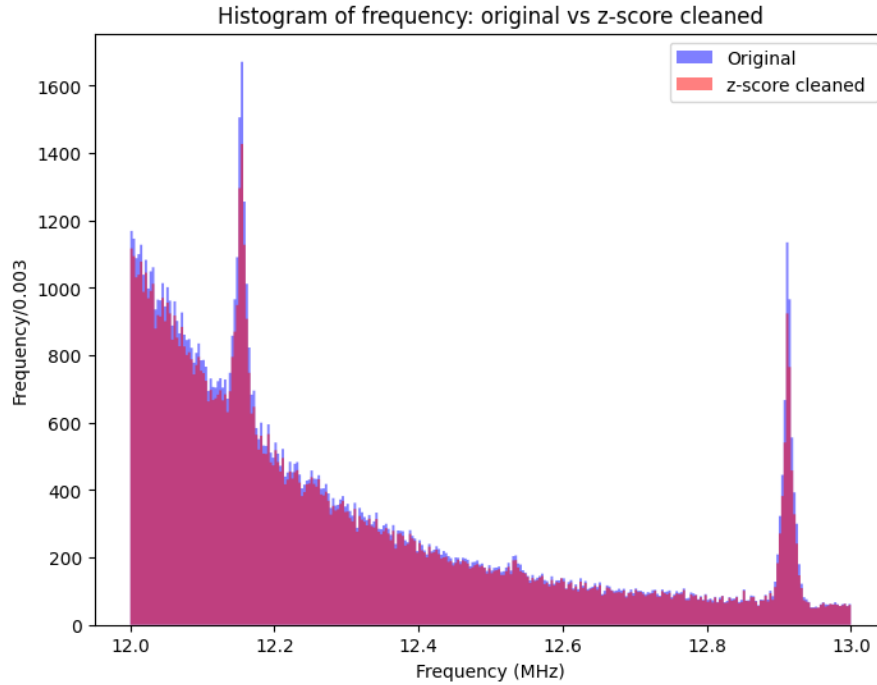


Figure 29: Histogram of original data and z-score cleaned data. Poisson errors are omitted for visualization.

As can be seen on fig. 29, not much has changed.

Using the z-score cleaned data, and refitting peak 1 and 2, again with the Cauchy and exponential background model, we get, for the first peak:

And for peak 2 (fig. 31):

Looking at both fig. 30 and fig. 31 and comparing to fig. 24 and fig. 26 we don't see much of a difference w.r.t. χ^2 and p-values. This is why we chose the second approach as well.

Fishers LDA approach

Looking at fig. 21, and using our knowledge about peak positions, we separate the distribution into signal and noise, as can be seen below (fig. 32):

Based on our given noise discriminating variables, D and E, we separate the signal and noise using the standard Fishers LDA approach, i.e. we loop over each specie (signal, noise) and each separation variable (D, E), and slowly build the covariance matrices. Combining and inverting the summed covariance matrix, we find the 2-component weight vector using the difference of the mean vectors for each specie based on D and E. We find:

$$\omega = [-0.15, 0.11] \quad (30)$$

Using eq. (30), we find the distributions of Fisher values for each signal and noise to be:

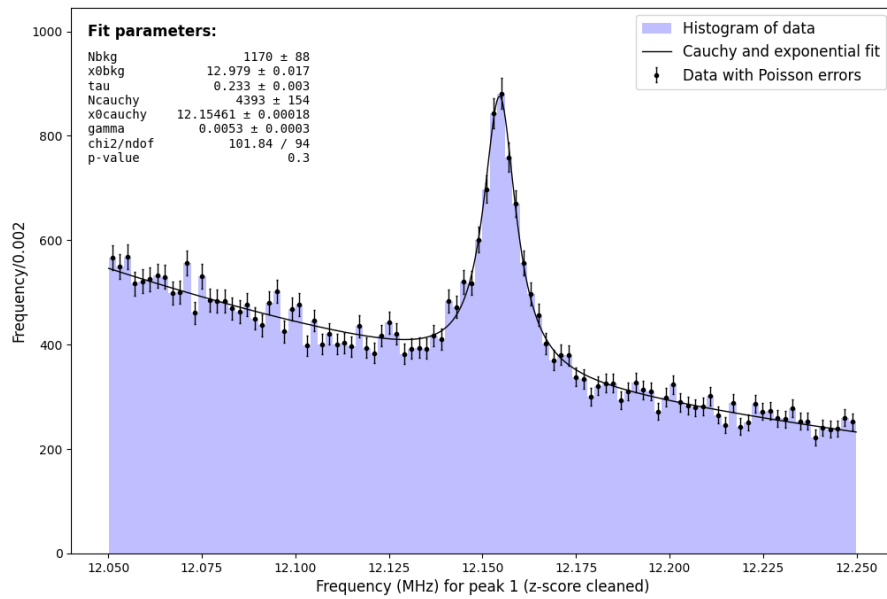


Figure 30: Fit of peak 1 with Cauchy plus exponential background model on z-score cleaned data.

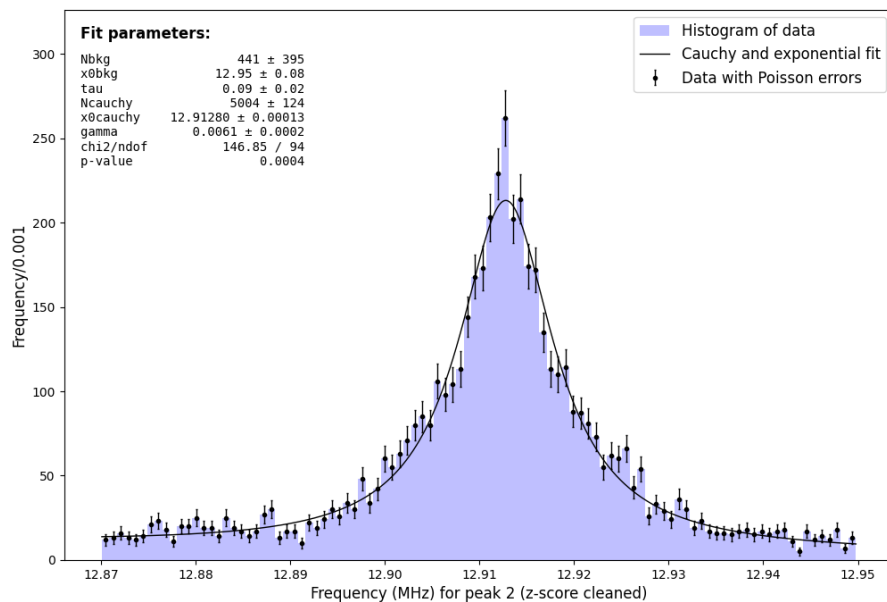


Figure 31: Fit of peak 2 with Cauchy plus exponential background model on z-score cleaned data.

The separation found is $\Delta = 0.41$. Based on this, we extract only frequencies whose LDA score is above 90%.

Plotting the original and LDA-cleaned data together, we find:

Much better cleaning!

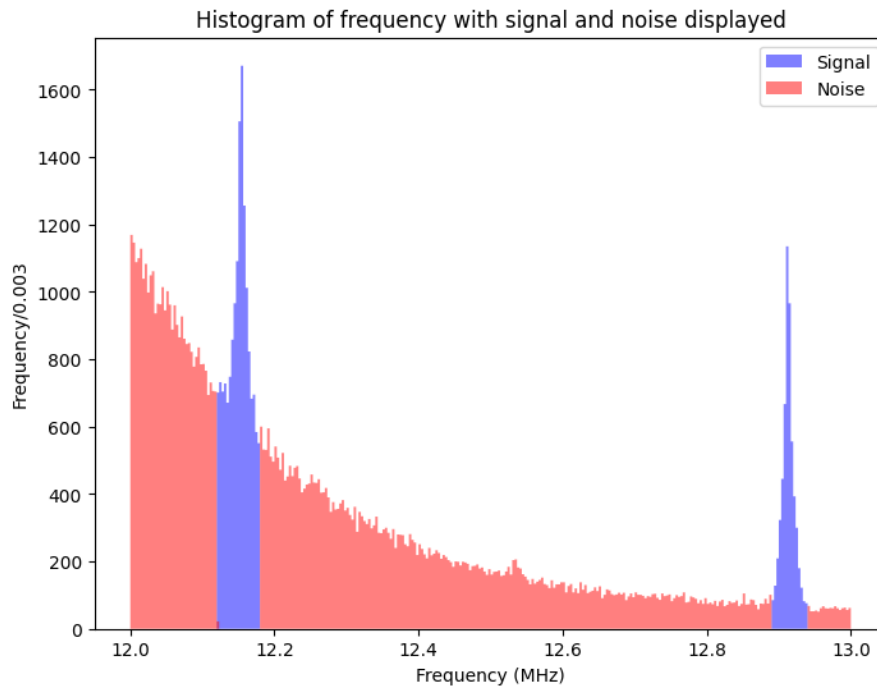


Figure 32: Divide between signal and noise in distribution of frequencies. Poisson errors are omitted for visualization.

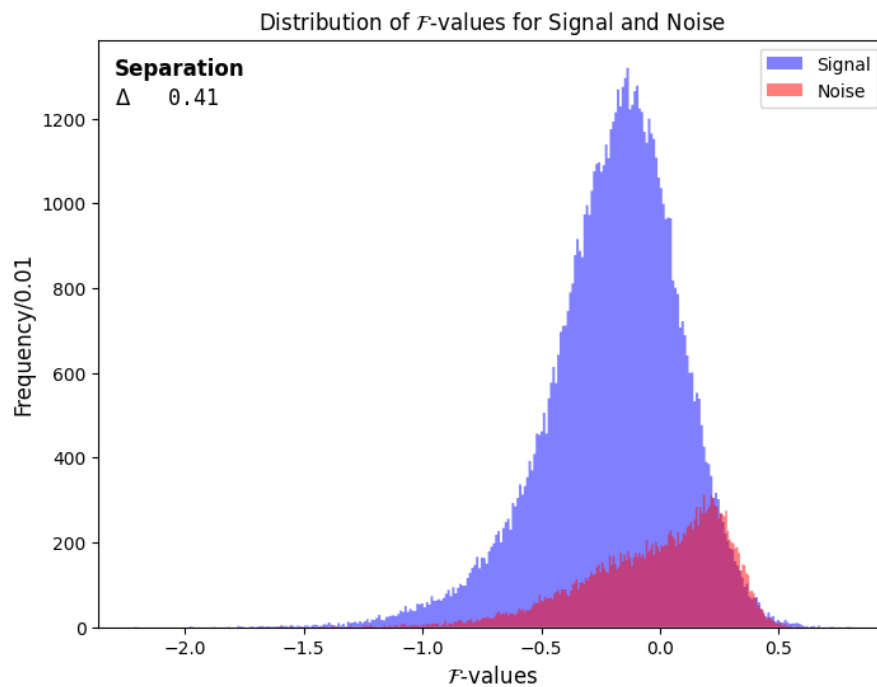


Figure 33: Distribution of Fisher values for each specie (signal, noise)

Refitting the cleaned data showed that the Cauchy and exponential background model is not very good. Instead, we've fitted with a double Gaussian model.

For the first peak we find:

Looking at fig. 35, we see a very good fit. The χ^2/ndof value is not a lot higher than 1, and the p-value is $p = 0.10$, quite above a reasonable 5% significant level. This is good!

For the second peak we find:

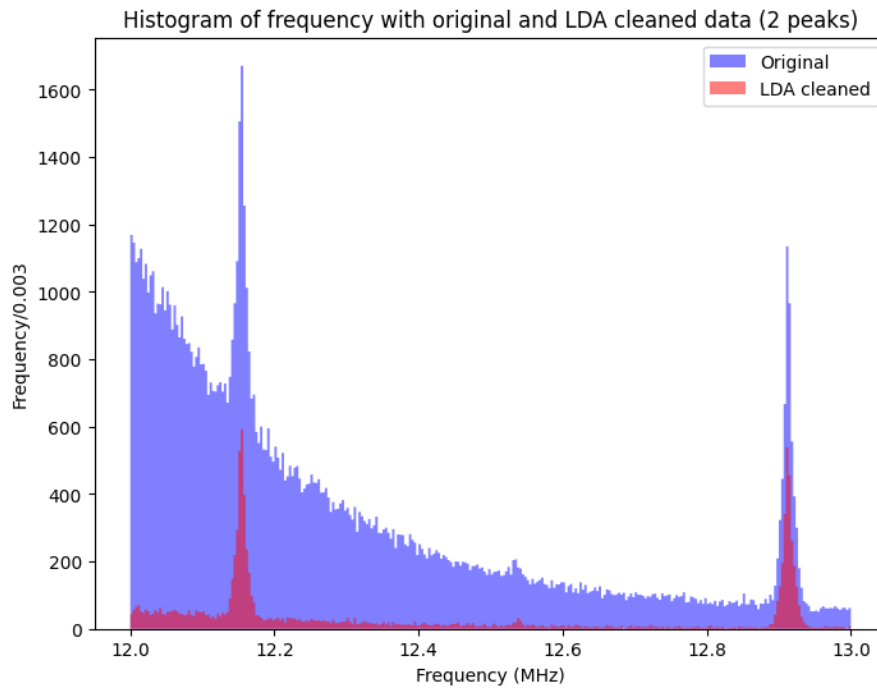


Figure 34: Original and cleaned frequency data using LDA. Poisson errors are omitted for visualization.

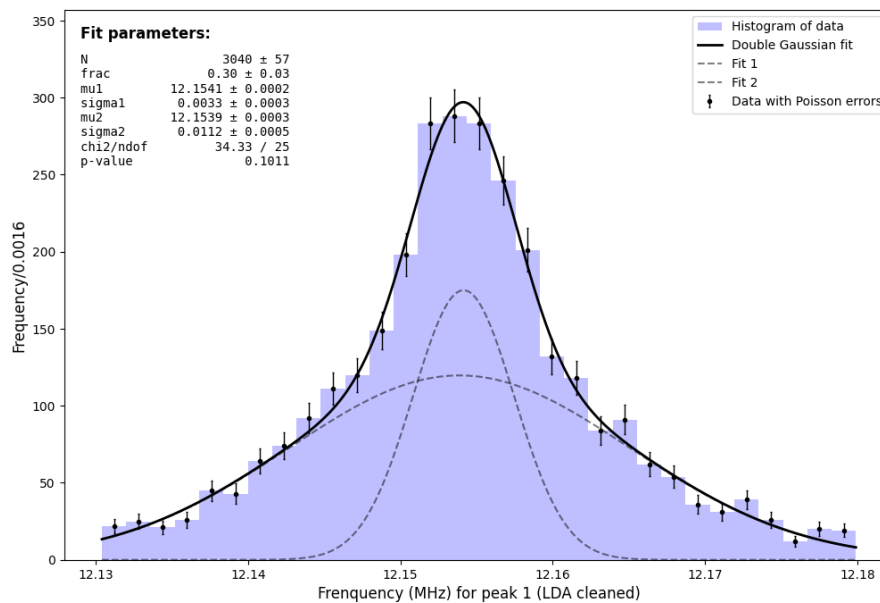


Figure 35: Double Gaussian fit on first peak using LDA-cleaned data.

Looking at fig. 36, we again see a good fit. Here χ^2 and p-values are good, about the same as in fig. 35.

5.2.4

Looking at fig. 21 there is a small peak at about 12.5 (MHz). Lets try and fit the third peak (small one). Our approach is the same as the LDA approach above. But, based on trial and error, we found the prettiest fit using a 80% threshold instead of 90%, otherwise the peak is almost gone.

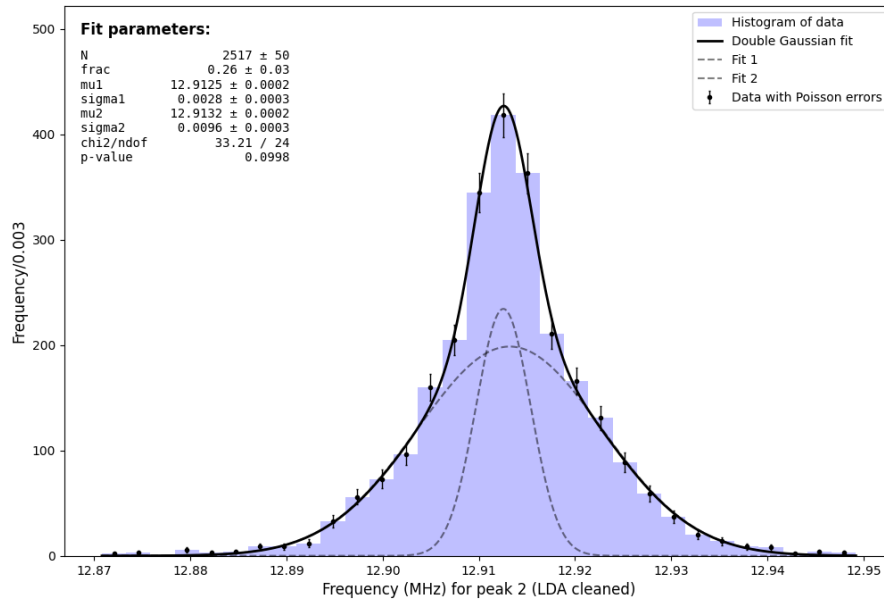


Figure 36: Double Gaussian fit on second peak using LDA-cleaned data.

We now have:

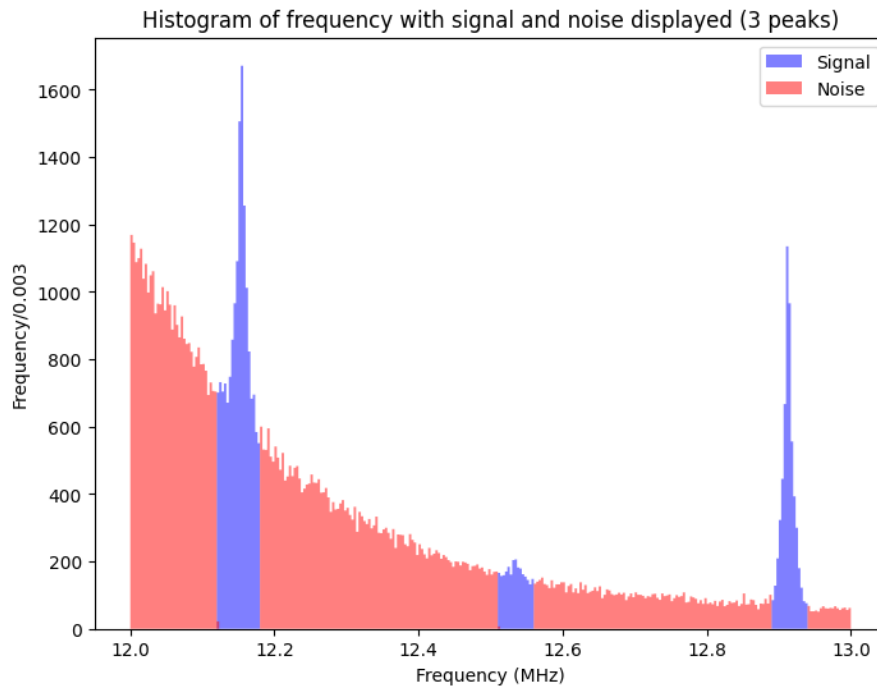


Figure 37: Histogram showcasing the choice of signal and noise when trying to incorporate the third peak. Poisson errors are omitted for visualization.

We now obtain:

$$\omega_{3p} = [-0.136, 0.100] \quad (31)$$

This gives us the following distribution of Fishers discriminant values:

Now obtaining a smaller separation of $\Delta_{3p} = 0.37$, since we have more noise in our signal group. Cleaning the data with 80% cut-off, we get:

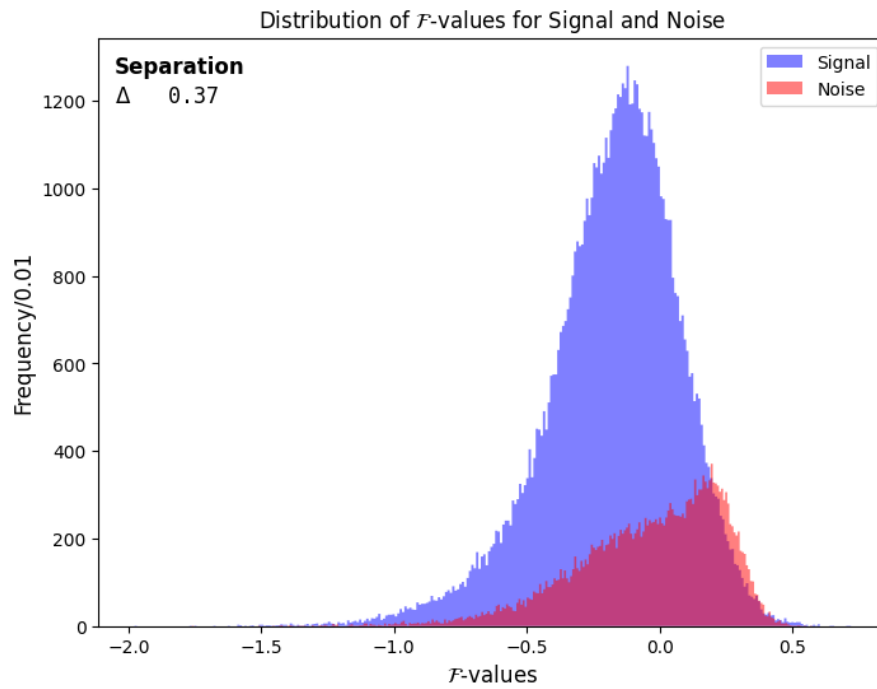


Figure 38: Distribution of Fisher values for each specie (signal, noise) with third peak.

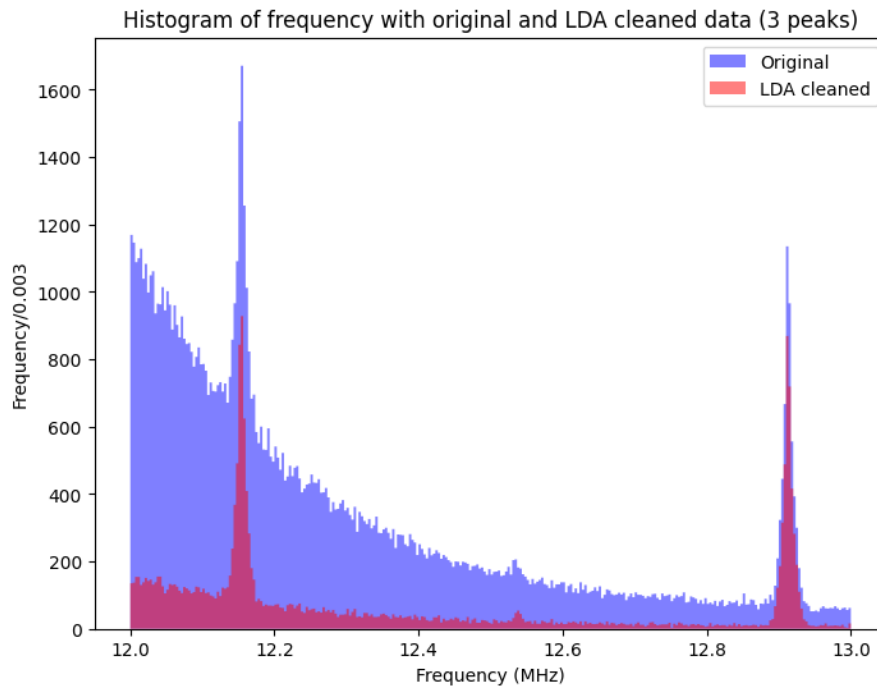


Figure 39: Original and cleaned frequency data using LDA with third peak.

A pretty good clean in my opinion. Zooming in on the third peak (small one) and fitting with a double Gaussian, we obtain:

Looking at fig. 40, we have made a pretty good fit. The χ^2/ndof -value is only a bit smaller than 1, and the p-value found is $p = 0.4308$, just smack in the middle of 0 and 1, and thus far away from any reasonable significant level. So we do get pretty significant!

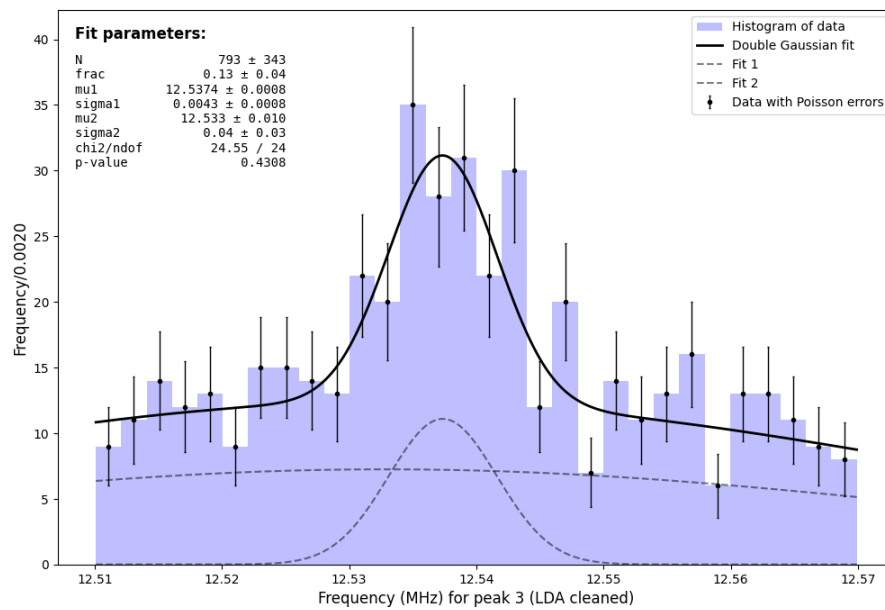


Figure 40: Double Gaussian fit on third peak (small one) using LDA cleaned data. Poisson errors are omitted for visualization

End of my solution.

UCPH's AI declaration

Declaration of using generative AI tools

☒ I/we have used generative AI as an aid/tool *(please tick)*

☐ I/we have NOT used generative AI as an aid/tool *(please tick)*

If generative AI is permitted in the exam, but you haven't used it in your exam paper, you just need to tick the box stating that you have not used GAI. You don't have to fill in the rest.

List which GAI tools you have used and include the link to the platform (if possible):

GitHub Copilot

Describe how generative AI has been used in the exam paper:

Github Copilot used in accordance with the exam rules for AppStat.